Byung-Hak Hwang

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Research interests

- AI for mathematics.
- Formalization of mathematics.
- Algebraic combinatorics and related topics.

Education and Employment

- **AI Fellow, Center for Artificial Intelligence and Natural Sciences, Korea Institute for Advanced Study**
  March 2026 –
- **Research Fellow, Korea Institute for Advanced Study**
  May 2024 – February 2026
  Mentor: Hyun Kyu Kim
- **Alternative Military Service**
  April 2021 – April 2024
- **Research Fellow, Applied Algebra and Optimization Research Center**
  September 2020 – March 2021
  Mentor: Jang Soo Kim
- **Ph.D. Mathematics, Seoul National University**
  March 2014 – August 2020
  Advisor: Woong Kook
- **B.S. Double major in Mathematics, and Electric & Computer Engineering, Seoul National University**
  March 2009 – February 2014

Publications

Journal publications and preprints

11. *Chromatic functions of interval graphs and Ψ-positivity* (with M. D'Adderio, G. Interdonato)
    preprint, submitted.

10. *Composition Direction of Seymour's Theorem for Regular Matroids – Formally Verified* (with M. Dvorak, T. Figueroa-Reid, R. Hamadani, E. Karunus, V. Kolmogorov, A. Meiburg, A. Nelson, P. Nelson, M. Sandey, I. Sergeev)
    preprint, arXiv:2509.20539, submitted.

9. *Refinement of Hikita's e-positivity theorem via Abreu–Nigro's g-functions and restricted modular law* (with J. Huh, D. Kim, J. S. Kim, J. Oh)
   preprint, arXiv:2504.09123, submitted.

8. *Refined canonical stable Grothendieck polynomials and their duals, Part 2* (with J. Jang, J. S. Kim, M. Song, U-K. Song)
   European Journal of Combinatorics, 127:Paper No. 104166, 34, 2025.

7. *Noncommutative symmetric functions and skewing operators*
   Discrete Mathematics, 348(1):Paper No. 114255, 2025.
   Editors' Choice Award 2025

6. *Refined canonical stable Grothendieck polynomials and their duals, Part 1* (with J. Jang, J. S. Kim, M. Song, U-K. Song)
   Advances in Mathematics, 446:Paper No. 109670, 42, 2024.

5. *Chromatic quasisymmetric functions and noncommutative P-symmetric functions*
   Transactions of the American Mathematical Society, 377(4):2855-2896, 2024.

4. *A combinatorial model for the transition matrix between the Specht and SL_2-web bases* (with J. Jang and J. Oh)
   Forum of Mathematics, Sigma, 11 (2023), E82.

3. *Acyclic orientation polynomials and the sink theorem for chromatic symmetric functions* (with W.-S. Jung, K.-J. Lee, J. Oh, and S.-H. Yu)
   Journal of Combinatorial Theory, Series B, 149 (2021), 52-75.

2. *On linearization coefficients of q-Laguerre polynomials* (with J. S. Kim, J. Oh, and S.-H. Yu)
   The Electronic Journal of Combinatorics, Volume 27, Issue 2 (2020), P2.22.

1. *Reverse plane partitions of skew staircase shapes and q-Euler numbers* (with J. S. Kim, M. Yoo, and S.-m. Yun)
   Journal of Combinatorial Theory, Series A, 168 (2019), 120-163.

Conference proceedings

6. *Refined canonical stable Grothendieck polynomials and their duals* (with J. S. Kim, J. Jang, M. Song, and U-K. Song)
   Proceedings of FPSAC 2023.

5. *Chromatic quasisymmetric functions and noncommutative P-symmetric functions*
   Proceedings of FPSAC 2023.

4. *A combinatorial model for the transition matrix between the Specht and web bases* (with J. Jang, and J. Oh)
   Proceedings of FPSAC 2022.

3. *Acyclic orientation polynomials and the sink theorem for chromatic symmetric functions* (with W.-S. Jung, K.-J. Lee, J. Oh, and S.-H. Yu)
   Proceedings of FPSAC 2020.

2. *On linearization coefficients of q-Laguerre polynomials* (with J. S. Kim, J. Oh, and S.-H. Yu)
   Proceedings of FPSAC 2020.

1. *Reverse plane partitions of skew staircase shapes and q-Euler numbers* (with J. S. Kim, M. Yoo, and S.-m. Yun)
   Proceedings of FPSAC 2018.

Honors and Awards

- Editors' Choice Award 2025, Discrete Mathematics for the paper "Noncommutative symmetric functions and skewing operators"
- Outstanding Teaching Assistant Award, Department of Mathematical Sciences, Seoul National University, 2015

Teaching and Mentoring

- Lecturer and Mentor, KRIMS 2026 Winter School—Symmetric Functions and Group Representations, February 2026
- Mentor, 2025 KIAS Winter School on Mathematics and AI, December 2025
- Teaching Assistant, Department of Mathematical Sciences, Seoul National University (2014-2020)

Colloquia and Invited Talks

- Refinement of Hikita's e-positivity theorem via Abreu–Nigro's g-functions and restricted modular law, AlCoVE 2025, Invited talk, May 2025
- Formalizing mathematics: why and how, Department of Mathematics, Korea University, Invited talk, May 2025
- Formalizing mathematics: why and how, Department of Mathematics, Sungkyunkwan University, Colloquium, March 2025
- Braid varieties, R-polynomials, and rational Catalan number, Combinatorics on flag varieties and related topics 2025, Invited talk, February 2025
- Formalizing mathematics: why and how, AI for Mathematics Workshop on Formalization, KIAS, Invited talk, December 2024
- Formalizing mathematics: why and how, Department of Mathematics, Hanyang University, Colloquium, December 2024
- Chromatic quasisymmetric functions and noncommutative P-symmetric functions, Combinatorics on flag varieties and related topics 2023, Invited talk, February 2023

Contributed Talks

- Formalizing mathematics: why it matters now, KIAS Center for AI and Natural Sciences 2026 Winter Workshop, January 2026
- Formalizing mathematics: why and how, Rookies workshop 2025, Seoul National University, November 2025
- Formalizing mathematics: why and how, HCMC Quantum Analysis & Geometry, August 2025
- Formalizing mathematics: why and how, 2025 KMS Annual Meeting, KAIST, April 2025
- Noncommutative symmetric functions and skewing operators, The 33rd KIAS Combinatorics Workshop, KIAS, February 2025

- Formalizing mathematics: why and how, Combinatorics Seminar for Young Researchers, Inha University, January 2025
- Cluster combinatorics of Grassmannians, Intensive workshop on algebraic combinatorics, January 2025
- Braids and R-polynomials, 2024 KMS Annual Meeting, Sungkyunkwan University, October 2024
- Formalizing mathematics: why and how, Workshop on Mathematical Challenges 2024, KIAS, October 2024
- The theory of noncommutative symmetric functions, Topology and Combinatorics seminar at Ajou University, June 2024
- Logarithmic concavity of Kazhdan–Lusztig $\widetilde{R}$-polynomials, Workshop for Young Representation Theorist in Korea, December 2023
- Chromatic quasisymmetric functions and noncommutative P-symmetric functions, FPSAC 2023, July 2023
- A proof of Harada–Precup's conjecture, SKKU Combinatorics seminar, November 2020
- Acyclic orientation polynomials, 2020 KMS Annual Meeting, October 2020
- Positivity in symmetric functions, AORC Monthly seminar, September 2020
- Chromatic quasisymmetric functions and noncommutative P-symmetric functions, Topology and Combinatorics seminar at Ajou University, September 2020
- Acyclic orientation polynomials, 2020 Combinatorics Workshop, August 2020
- Acyclic orientation polynomials and the sink theorem for chromatic symmetric functions (poster), FPSAC 2020 (online), July 2020
- Reverse plane partitions of skew staircase shapes and a modification of Lindstörm–Gessel–Viennot lemma, The 18th KIAS Combinatorics Workshop, 2017